Student Exam Management System

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Project Type: C# Console Application

1. Project Overview

The **Student Exam Management System** is a console application built in C# for managing **students, teachers, exams, and exam results**.

The system stores data in JSON files and provides features for:

- Adding and managing records.
- Validating inputs (IDs, GPA, Emails, Dates).
- Reporting student results and top performers.
- Exporting exam results to CSV for further analysis.

2. Features & Main Menu

The application provides a main menu with the following options:

Option	Functionality
1	Add Student
2	Add Teacher
3	Show/Search/Sort Students
4	Add Exam
5	Register Exam Result
6	Show Student Results
7	Show Top 3 Students in a Subject
8	Save Data

9	Load Data
10	Export Results to CSV
11	Exit
12	Update Student
13	Delete Student

3. How to Run

1. Open **Command Prompt** (Windows) or **Terminal** (Mac/Linux).
2. Navigate to the project folder.
3. Run the project using:

```
dotnet run
```

4. Follow the **Main Menu prompts** to use the application.

4. Data Storage

All data is stored in **JSON files** inside the `Data` folder:

File	Description
`students.json`	Stores student information
`teachers.json`	Stores teacher information
`exams.json`	Stores exam information
`results.json`	Stores exam results
`results.csv`	Exported exam results for reporting

5. Validations

- **IDs** must be numeric and unique.
- **Emails** are validated using Regex to ensure proper format.
- **GPA** must be between `0` and 4.

- **Exam dates** must follow `yyyy-MM-dd` format.
- **Scores** must be positive numbers.

6. Sample Usage

- **Add a Student** → Enter ID, Name, Email, Level, GPA.
- **Update a Student** → Modify name, email, level, or GPA.
- **Delete a Student** → Remove student record by ID.
- **Add a Teacher** → Enter ID, Name, Email, Subject.
- **Add an Exam** → Enter ID, Subject, and Date.
- **Register a Result** → Enter Student ID, Exam ID, and Score.
- **Show Student Results** → View all exams and scores for a student.
- **Show Top 3 Students** → Enter a subject to see top performers (average scores).
- **Export Results to CSV** → Generate `results.csv` in the Data folder.

7. Dependencies

- **.NET 6.0** or higher
- `System.Text.Json` → JSON serialization/deserialization
- `System.IO` → CSV export
- **LINQ** → filtering, grouping, and ordering data

8. Notes

- Ensure **IDs are unique**; duplicates will be rejected.
- The **Data folder must be writable**.
- CSV export **overwrites existing `results.csv`**.
- Top 3 students report calculates the **average score** for all exams in a subject.